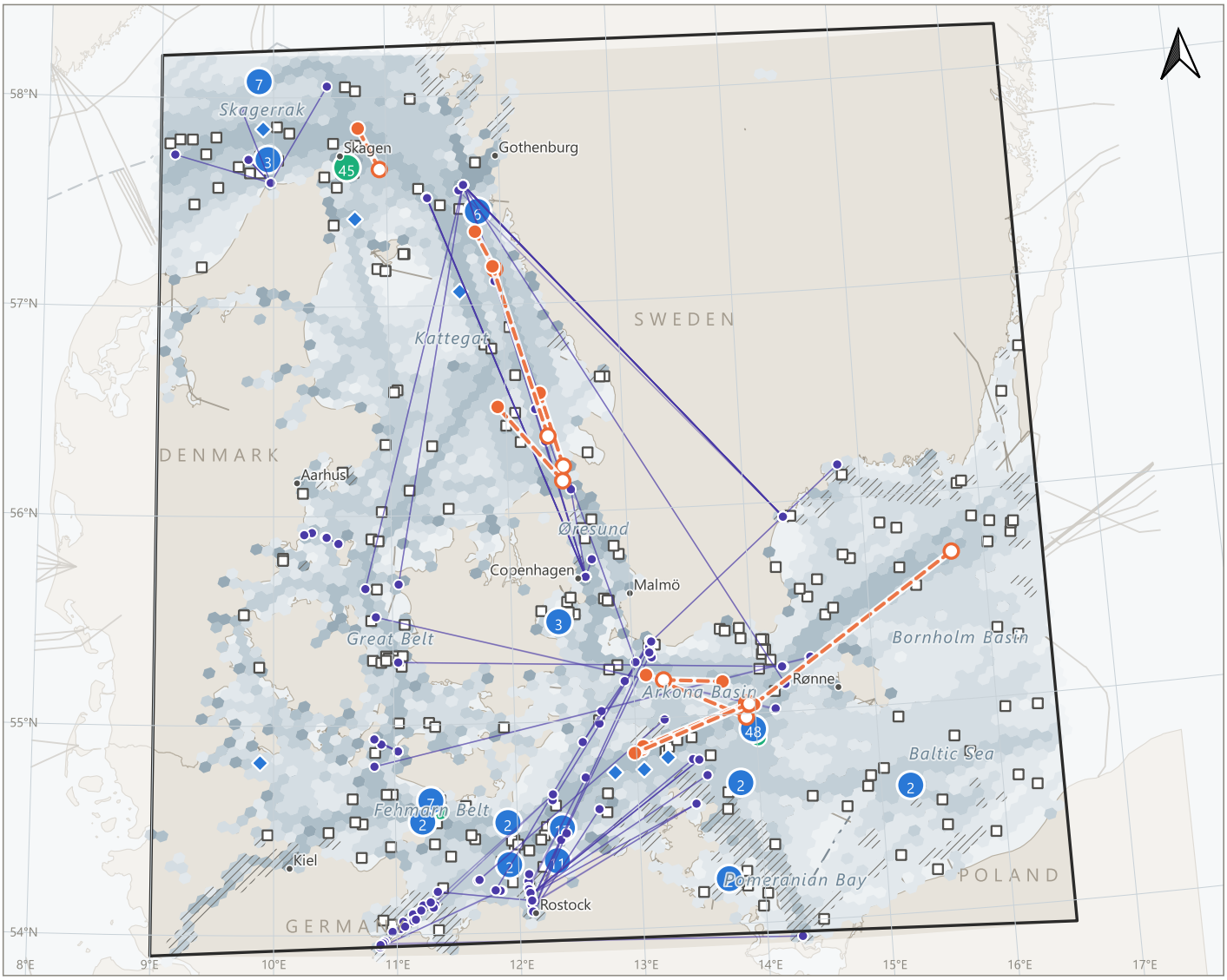
ZONE SHEET · SHORE-BASED AIS · OPEN DATA

What AIS doesn't show

Danish Straits & Bornholm Basin · Study area · 13–19 October 2026



0 30 60 NM

Projection ETRS89-LAEA (EPSG:3035). Grey shading: shipping density.

KEY FIGURES

2,894
class A vessels tracked

3,880
AIS silences of 2 hours or more

496
silences explained by shore reception gaps

11
unexplained: silent transits in well-covered waters

91 %
of underway traffic in reliably covered waters

210
radar detections without AIS (Sentinel-1, wind farms excluded)

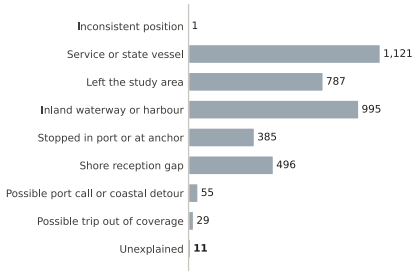
10
vessels whose MMSI is shared by two transmitters

121
slow passages over a cable or pipeline (30 min or more)

47
ship-to-ship encounters (2 hours or more)

94 %
of radar-seen vessels received by the shore AIS network

WHERE THE AIS SILENCES COME FROM



Unexplained AIS silence
MMSI shared by two transmitters
Slow passage over a cable or pipeline
Ship-to-ship encounter
Radar detection, no AIS
Cable or pipeline
Poor shore AIS reception
EEZ boundary

Method. Positions of class A vessels from the Danish shore AIS network, cleaned and thinned to one per minute. Shore coverage is measured from the traffic itself on an H3 grid, counting vessels lost while crossing each cell. A silence of 2 hours or more is called unexplained only after ruling out, in order: service or state vessel, exit from the area, inland waterway, port stay, reception gap, possible port call and possible trip out of coverage. Indicators flag deviations from expected behaviour, not offences; vessels are pseudonymised.

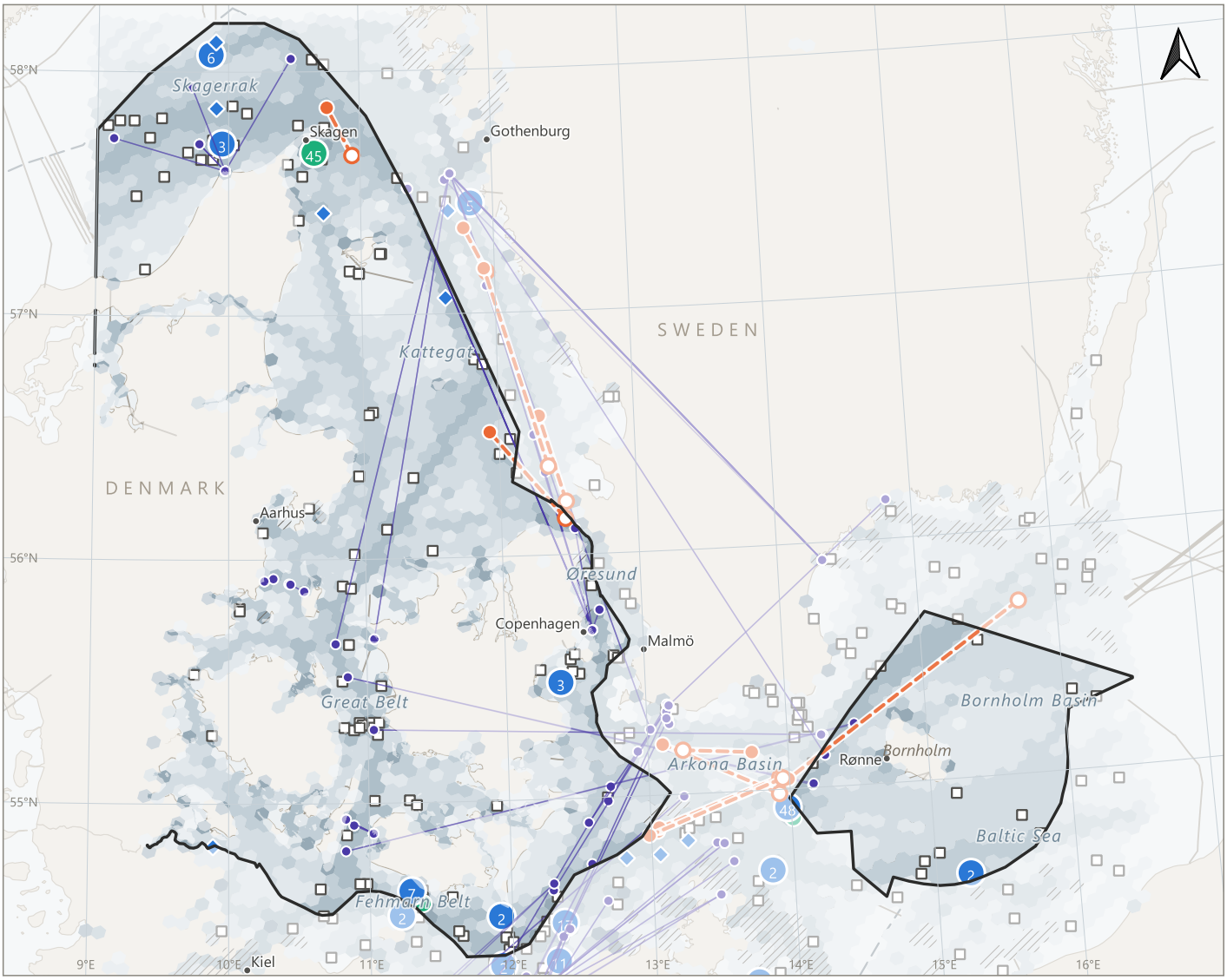
Data: Danish Maritime Authority (AIS) · Flanders Marine Institute, Marine Regions (EEZ, CC BY 4.0) · Eurostat GISCO (© EuroGeographics) · EMODnet Human Activities and BSH (CC BY 4.0) · TeleGeography (CC BY-NC-SA 3.0) · © OpenStreetMap contributors (ODbL) · Radar detections: Powered by Global Fishing Watch (CC BY-NC 4.0).

Independent portfolio project · Jaurès Daa-Hingbanon, GIS analyst

ZONE SHEET · SHORE-BASED AIS · OPEN DATA

What AIS doesn't show

Danish EEZ · Within the study area · 13–19 October 2026



0 20 40 NM

Projection ETRS89-LAEA (EPSG:3035). Grey shading: shipping density.

KEY FIGURES

2,002
class A vessels tracked

196
AIS silences of 2 hours or more

9
silences explained by shore reception gaps

2
unexplained: silent transits in well-covered waters

98 %
of underway traffic in reliably covered waters

99
radar detections without AIS (Sentinel-1, wind farms excluded)

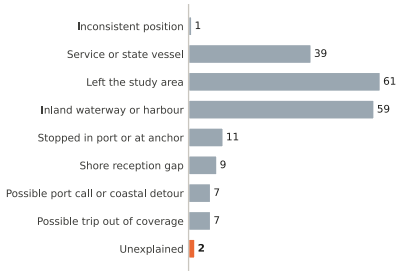
8
vessels whose MMSI is shared by two transmitters

32
slow passages over a cable or pipeline (30 min or more)

46
ship-to-ship encounters (2 hours or more)

100 %
of radar-seen vessels received by the shore AIS network

WHERE THE AIS SILENCES COME FROM



Unexplained AIS silence
MMSI shared by two transmitters
Slow passage over a cable or pipeline
Ship-to-ship encounter
Radar detection, no AIS
Cable or pipeline
Poor shore AIS reception
EEZ boundary

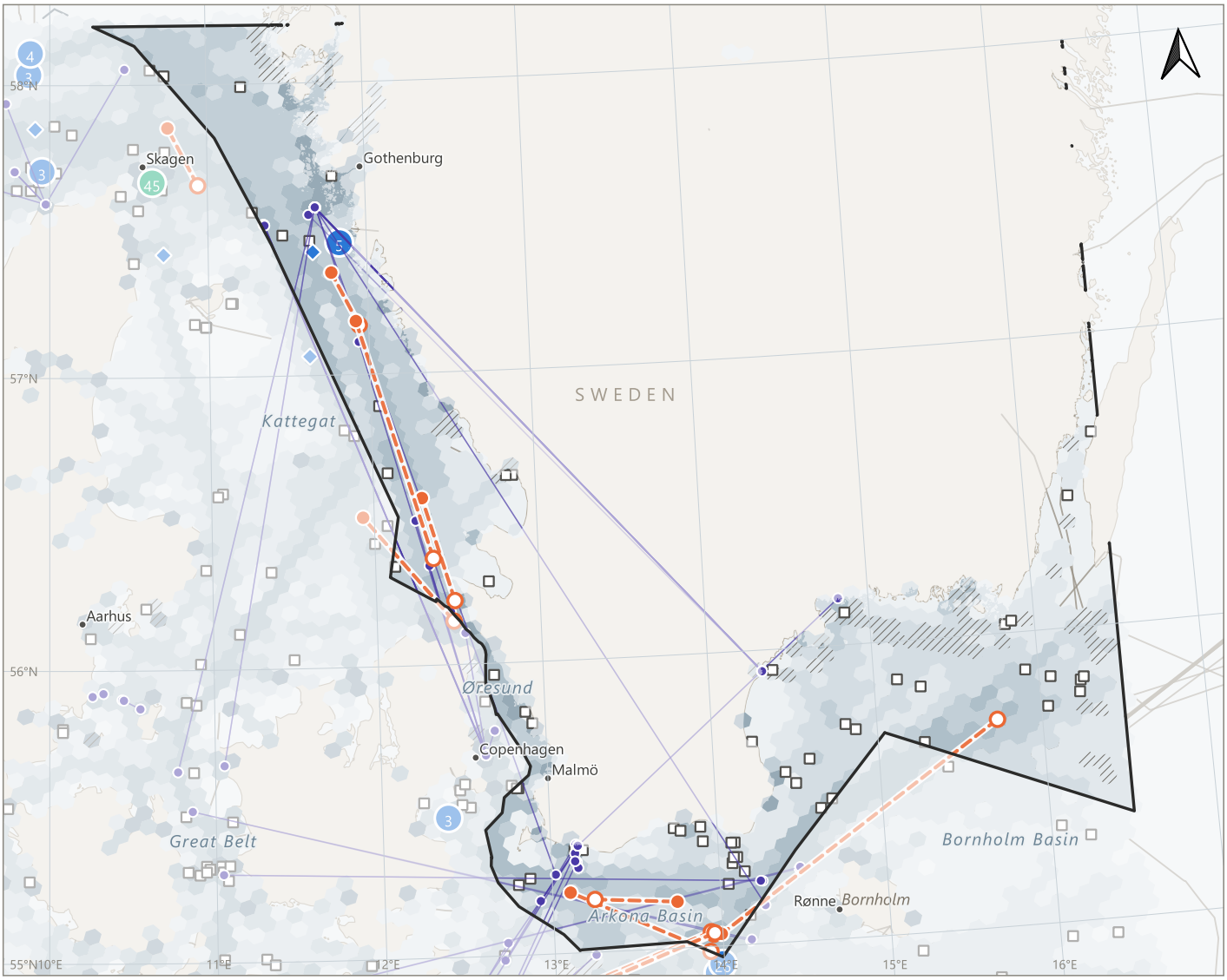
Method. Positions of class A vessels from the Danish shore AIS network, cleaned and thinned to one per minute. Shore coverage is measured from the traffic itself on an H3 grid, counting vessels lost while crossing each cell. A silence of 2 hours or more is called unexplained only after ruling out, in order: service or state vessel, exit from the area, inland waterway, port stay, reception gap, possible port call and possible trip out of coverage. Indicators flag deviations from expected behaviour, not offences; vessels are pseudonymised.

Data: Danish Maritime Authority (AIS) · Flanders Marine Institute, Marine Regions (EEZ, CC BY 4.0) · Eurostat GISCO (© EuroGeographics) · EMODnet Human Activities and BSH (CC BY 4.0) · TeleGeography (CC BY-NC-SA 3.0) · © OpenStreetMap contributors (ODbL) · Radar detections: Powered by Global Fishing Watch (CC BY-NC 4.0).

Independent portfolio project · Jaurès Daa-Hingbanon, GIS analyst

What AIS doesn't show

Swedish EEZ · Within the study area · 13–19 October 2026



0 20 40 NM

Projection ETRS89-LAEA (EPSG:3035). Grey shading: shipping density.

KEY FIGURES

1,429
class A vessels tracked

628
AIS silences of 2 hours or more

108
silences explained by shore reception gaps

9
unexplained: silent transits in well-covered waters

94 %
of underway traffic in reliably covered waters

55
radar detections without AIS (Sentinel-1, wind farms excluded)

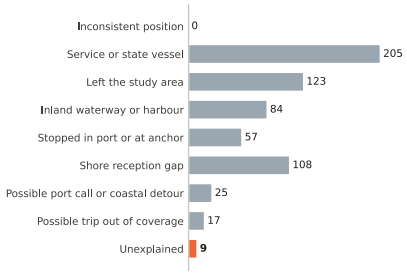
4
vessels whose MMSI is shared by two transmitters

6
slow passages over a cable or pipeline (30 min or more)

0
ship-to-ship encounters (2 hours or more)

93 %
of radar-seen vessels received by the shore AIS network

WHERE THE AIS SILENCES COME FROM



Unexplained AIS silence
MMSI shared by two transmitters
Slow passage over a cable or pipeline
Ship-to-ship encounter
Radar detection, no AIS
Cable or pipeline
Poor shore AIS reception
EEZ boundary

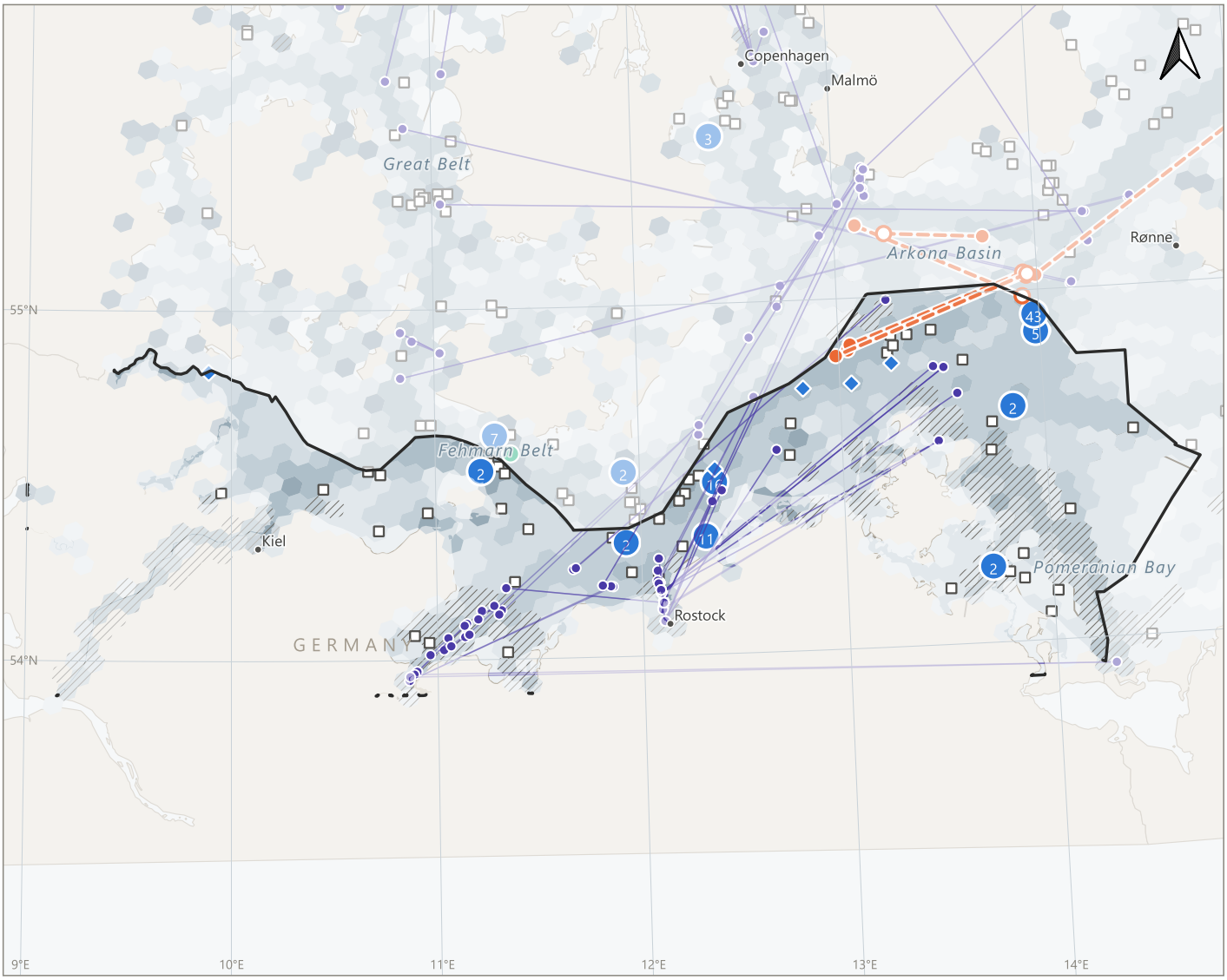
Method. Positions of class A vessels from the Danish shore AIS network, cleaned and thinned to one per minute. Shore coverage is measured from the traffic itself on an H3 grid, counting vessels lost while crossing each cell. A silence of 2 hours or more is called unexplained only after ruling out, in order: service or state vessel, exit from the area, inland waterway, port stay, reception gap, possible port call and possible trip out of coverage. Indicators flag deviations from expected behaviour, not offences; vessels are pseudonymised.

Data: Danish Maritime Authority (AIS) · Flanders Marine Institute, Marine Regions (EEZ, CC BY 4.0) · Eurostat GISCO (© EuroGeographics) · EMODnet Human Activities and BSH (CC BY 4.0) · TeleGeography (CC BY-NC-SA 3.0) · © OpenStreetMap contributors (ODbL) · Radar detections: Powered by Global Fishing Watch (CC BY-NC 4.0).

ZONE SHEET · SHORE-BASED AIS · OPEN DATA

What AIS doesn't show

German EEZ · Within the study area · 13–19 October 2026



0 20 40 NM

Projection ETRS89-LAEA (EPSG:3035). Grey shading: shipping density.

KEY FIGURES

1,203
class A vessels tracked

1,338
AIS silences of 2 hours or more

278
silences explained by shore reception gaps

0
unexplained: silent transits in well-covered waters

70 %
of underway traffic in reliably covered waters

44
radar detections without AIS (Sentinel-1, wind farms excluded)

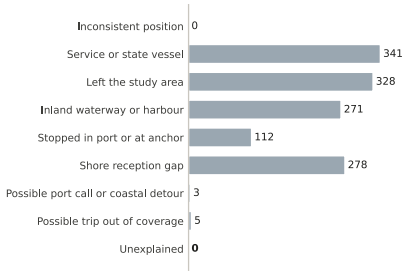
4
vessels whose MMSI is shared by two transmitters

83
slow passages over a cable or pipeline (30 min or more)

1
ship-to-ship encounters (2 hours or more)

89 %
of radar-seen vessels received by the shore AIS network

WHERE THE AIS SILENCES COME FROM



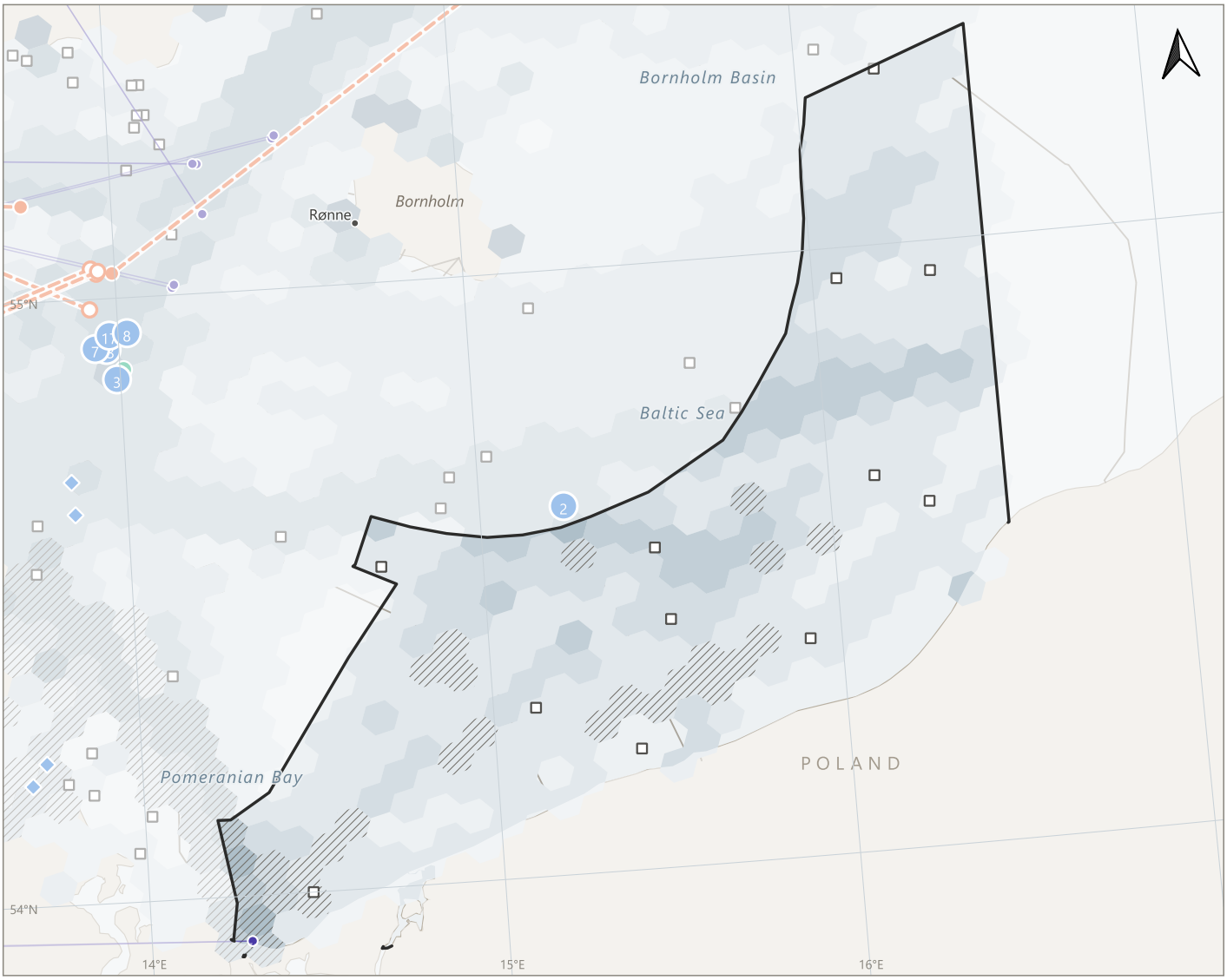
Unexplained AIS silence
MMSI shared by two transmitters
Slow passage over a cable or pipeline
Ship-to-ship encounter
Radar detection, no AIS
Cable or pipeline
Poor shore AIS reception
EEZ boundary

Method. Positions of class A vessels from the Danish shore AIS network, cleaned and thinned to one per minute. Shore coverage is measured from the traffic itself on an H3 grid, counting vessels lost while crossing each cell. A silence of 2 hours or more is called unexplained only after ruling out, in order: service or state vessel, exit from the area, inland waterway, port stay, reception gap, possible port call and possible trip out of coverage. Indicators flag deviations from expected behaviour, not offences; vessels are pseudonymised.

Data: Danish Maritime Authority (AIS) · Flanders Marine Institute, Marine Regions (EEZ, CC BY 4.0) · Eurostat GISCO (© EuroGeographics) · EMODnet Human Activities and BSH (CC BY 4.0) · TeleGeography (CC BY-NC-SA 3.0) · © OpenStreetMap contributors (ODbL) · Radar detections: Powered by Global Fishing Watch (CC BY-NC 4.0).

What AIS doesn't show

Polish EEZ · Within the study area · 13–19 October 2026



Projection ETRS89-LAEA (EPSG:3035). Grey shading: shipping density.

KEY FIGURES

281
class A vessels tracked

214
AIS silences of 2 hours or more

75
silences explained by shore reception gaps

0
unexplained: silent transits in well-covered waters

50 %
of underway traffic in reliably covered waters

12
radar detections without AIS (Sentinel-1, wind farms excluded)

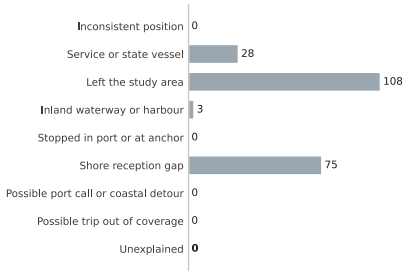
1
vessels whose MMSI is shared by two transmitters

0
slow passages over a cable or pipeline (30 min or more)

0
ship-to-ship encounters (2 hours or more)

52 %
of radar-seen vessels received by the shore AIS network

WHERE THE AIS SILENCES COME FROM



Unexplained AIS silence
MMSI shared by two transmitters
Slow passage over a cable or pipeline
Ship-to-ship encounter
Radar detection, no AIS
Cable or pipeline
Poor shore AIS reception
EEZ boundary

Method. Positions of class A vessels from the Danish shore AIS network, cleaned and thinned to one per minute. Shore coverage is measured from the traffic itself on an H3 grid, counting vessels lost while crossing each cell. A silence of 2 hours or more is called unexplained only after ruling out, in order: service or state vessel, exit from the area, inland waterway, port stay, reception gap, possible port call and possible trip out of coverage. Indicators flag deviations from expected behaviour, not offences; vessels are pseudonymised.

Data: Danish Maritime Authority (AIS) · Flanders Marine Institute, Marine Regions (EEZ, CC BY 4.0) · Eurostat GISCO (© EuroGeographics) · EMODnet Human Activities and BSH (CC BY 4.0) · TeleGeography (CC BY-NC-SA 3.0) · © OpenStreetMap contributors (ODbL) · Radar detections: Powered by Global Fishing Watch (CC BY-NC 4.0).